Lecture 3 Linear Subspaces & Linearly Independent Vectors

- Linear Subspace of $\mathbb{R}^n$

Subset D of $\mathbb{R}^n$:

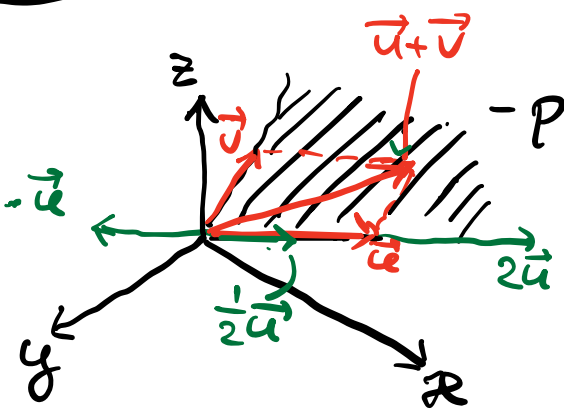
(i) $\vec{0} \in D$

(ii) For all $\vec{u}, \vec{v} \in D$: $\vec{u} + \vec{v} \in D$ (closed under addition)

(iii) For all $\vec{u} \in D$, $\alpha \in \mathbb{R}$: $\alpha \cdot \vec{u} \in D$ (closed under multiplication)

Important Message #1	Column-space and Null-space are linear subspaces

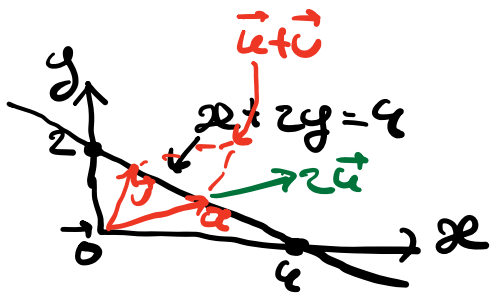
Example 1
$\mathbb{R}^3$



- plane, I only drew one quarter of it

← linear subspace of $\mathbb{R}^3$

Example 2
$\mathbb{R}^2$



(not linear subspace)

Definition: Linear Combination

$\vec{w}$ is a linear combination of $\vec{v}_1, \vec{v}_2, \ldots, \vec{v}_p$

$\Leftrightarrow$

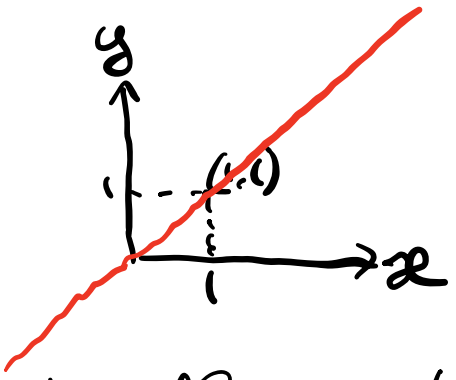
there are $c_1, c_2, \ldots, c_p \in \mathbb{R}$ such that

$$\vec{w} = c_1\vec{v}_1 + c_2\vec{v}_2 + \ldots + c_p\vec{v}_p$$

All linear combinations of $\vec{v}_1, \vec{v}_2, \ldots, \vec{v}_p$:

$$\text{Span}\{\vec{v}_1, \vec{v}_2, \ldots, \vec{v}_p\}$$
$$= \{\vec{w} \mid \vec{w} \text{ is a linear combination of } \vec{v}_1, \vec{v}_2, \ldots, \vec{v}_p\}$$

Example



$\text{Span}\left\{\begin{bmatrix}1\\1\end{bmatrix}\right\}$

Note: Span is always a linear subspace.

Example (Parametric Vector Form)

$$A\vec{x} = \vec{b} \qquad A = \begin{bmatrix} 3 & 5 & -4 \\ -3 & -2 & 4 \\ 6 & 1 & -8 \end{bmatrix} \qquad \vec{b} = \begin{bmatrix} 7 \\ -1 \\ -4 \end{bmatrix}$$

$$[A|\vec{b}] \sim \left[\begin{array}{ccc|c} 1 & 0 & -\frac{4}{3} & -1 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 0 & 0 \end{array}\right]$$

$$\left.\begin{array}{l} x_1 - \frac{4}{3}x_3 = -1 \\ x_2 = 2 \\ x_3 \text{ free variable} \end{array}\right\} \Rightarrow \begin{array}{l} x_1 = -1 + \frac{4}{3}x_3 \\ x_2 = 2 + 0 \cdot x_3 \\ x_3 = 0 + 1 \cdot x_3 \end{array}$$

$$\vec{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} -1 + \frac{4}{3}x_3 \\ 2 + 0 \cdot x_3 \\ 0 + 1 \cdot x_3 \end{bmatrix} = \begin{bmatrix} -1 \\ 2 \\ 0 \end{bmatrix} + \begin{bmatrix} \frac{4}{3}x_3 \\ 0 \cdot x_3 \\ 1 \cdot x_3 \end{bmatrix} =$$

$$\begin{bmatrix}-1\\2\\0\end{bmatrix} + \begin{bmatrix}\frac{4}{3}\\0\\1\end{bmatrix} \cdot x_3 = \begin{bmatrix}-1\\2\\0\end{bmatrix} + \text{Span}\left\{\begin{bmatrix}\frac{4}{3}\\0\\1\end{bmatrix}\right\}$$

Definition: A $k \times n$ matrix $[\vec{a}_1 \ \vec{a}_2 \ \dots \ \vec{a}_n]$

Column-space: Col A

$$= \text{Span}\{\vec{a}_1, \vec{a}_2, \dots, \vec{a}_n\}$$

$$= \{\vec{b} \in \mathbb{R}^k \mid A\vec{x} = \vec{b} \text{ for some } \vec{x}\}$$

$$[\vec{a}_1 \ \vec{a}_2 \ \dots \ \vec{a}_n] \begin{bmatrix}x_1\\x_2\\\vdots\\x_n\end{bmatrix} = \vec{a}_1 \cdot x_1 + \vec{a}_2 x_2 + \dots + \vec{a}_n x_n$$

Null-Space: $\text{Null } A = \{\vec{x} \in \mathbb{R}^n \mid \underbrace{A\vec{x} = 0}_{\text{homogeneous}}\}$

Example: $A\vec{x} = \vec{b}$ $\quad \vec{b} = \begin{bmatrix}7\\-1\\-4\end{bmatrix}$

is $\vec{b} \in \text{Col } A$? Yes, since $A\vec{x} = \vec{b}$ is consistent.

Linearly Independent Vectors

$\{\vec{v}_1, \vec{v}_2, \dots, \vec{v}_p\}$ are linearly independent if and only if

$x_1\vec{v}_1 + x_2\vec{v}_2 + \dots + x_p\vec{v}_p = \vec{0}$ only has solution $x_1 = x_2 = \dots x_p = 0$

Otherwise the vector set is linearly dependent.

Subspace $V = \text{span}\{\vec{v}_1, \vec{v}_2, \ldots, \vec{v}_p\}$

we want to describe V with the least amount of vectors possible.

$\{\vec{w}_1, \vec{w}_2, \ldots, \vec{w}_k\} \subseteq V$ form a **basis** of V if:

(1) $\{\vec{w}_1, \vec{w}_2, \ldots, \vec{w}_k\}$ are linearly independent

(2) $\text{Span}\{\vec{w}_1, \vec{w}_2, \ldots, \vec{w}_k\} = V$

Note that the basis needs not be unique (but # of (nonzero) vectors is) ← dimension

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Example: $\mathbb{R}^2$: $\left\{\begin{bmatrix}1\\0\end{bmatrix}, \begin{bmatrix}0\\1\end{bmatrix}\right\}$

$$\begin{bmatrix}x\\y\end{bmatrix} = x\begin{bmatrix}1\\0\end{bmatrix} + y\begin{bmatrix}0\\1\end{bmatrix} = \begin{bmatrix}x\\0\end{bmatrix} + \begin{bmatrix}0\\y\end{bmatrix} = \begin{bmatrix}x\\y\end{bmatrix}$$

but $\left\{\begin{bmatrix}1\\1\end{bmatrix}, \begin{bmatrix}2\\1\end{bmatrix}\right\}$ is also a basis.

Example: A : $m \times n$ matrix

dim Null A : Number of free variables ($A\vec{x} = 0$)

dim Col A : Number of basic variables

Since n columns → n variables in total

dim Null A + dim Col $A = n$

A : $n \times n$ (square) matrix. The following are equivalent:

- A is invertible
- Null $A = \{\vec{0}\}$ (dim Null $A = 0$)
- Col $A = \mathbb{R}^n$ (dim Col $A = n$)

Important Message #2

A $(n \times n)$ is invertible if and only if all columns are linearly independent

Theorem (3.17 LN): $\{\vec{v}_1, \ldots, \vec{v}_p\}$ basis of $V \subseteq \mathbb{R}^n$

- $p \leq n$
- if $p = n$ the $V = \mathbb{R}^n$
- if $V = \text{Span}\{\vec{w}_1, \vec{w}_2, \ldots \vec{w}_q\}$ $p \leq q$ and if $p = q$ then $\{\vec{w}_1, \vec{w}_2, \ldots \vec{w}_q\}$ are also linearly independent.

How to find a basis

For coefficient matrix A:

- elementary <u>row</u> operations do not affect the <u>solution set</u> of the associated linear system (but do change $\text{Col}\, A$)
- elementary <u>columns</u> operations do not affect <u>Col A</u> (but do change the solution set of the associated linear system)

Example $\text{Span}\{\vec{w}_1, \vec{w}_2, \vec{w}_3\} = V$

$$\vec{w}_1 = \begin{bmatrix} 1 \\ -2 \\ 4 \end{bmatrix}, \quad \vec{w}_2 = \begin{bmatrix} 4 \\ -7 \\ 9 \end{bmatrix}, \quad \vec{w}_3 = \begin{bmatrix} -4 \\ 10 \\ -30 \end{bmatrix}$$

① Construct matrix with vectors in V as columns.

② Compute an echelon form via column operations.

$$A = (\vec{w}_1 \ \vec{w}_2 \ \vec{w}_3) = \begin{bmatrix} 1 & 4 & -4 \\ -2 & -7 & 10 \\ 4 & 9 & -30 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & -4 \\ -2 & 3 & 10 \\ 4 & -7 & -30 \end{bmatrix}$$

$+1$ $\quad$ $\times 4$

$$\rightarrow \begin{bmatrix} 1 & 0 & 0 \\ -2 & 3 & 2 \\ 4 & -7 & -14 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & 0 \\ -2 & 6 & 6 \\ 4 & -42 & -42 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & 0 \\ -2 & 6 & 0 \\ 4 & -42 & 0 \end{bmatrix}$$

$\times 2$ $\quad$ $\times 3$

So 3rd column is dependent on 1st and 2nd column.

$$\text{Span}\{\vec{w}_1, \vec{w}_2, \vec{w}_3\} = \text{Span}\left\{\begin{bmatrix} 1 \\ -2 \\ 4 \end{bmatrix}, \begin{bmatrix} 0 \\ 6 \\ -42 \end{bmatrix}\right\}$$

Example

$\vec{w}_3$ is linear combination of $\vec{w}_1$ & $\vec{w}_2$

$$\vec{w}_3 = c_1\vec{w}_1 + c_2\vec{w}_2$$

(c_1, c_2) : coordinate vector of $\vec{w}_3$ with respect to basis $B = \{\vec{w}_1, \vec{w}_2\}$

$$= [\vec{w}_3]_B$$

Let's find the coordinates:

$$\vec{w}_3 = c_1\vec{w}_1 + c_2\vec{w}_2 = [\vec{w}_1 \ \vec{w}_2] \begin{bmatrix} c_1 \\ c_2 \end{bmatrix}$$

Augmented matrix (using elementary row operations)

$$\left[\begin{array}{cc|c} 1 & 4 & -4 \\ -2 & -7 & 10 \\ 4 & 9 & -30 \end{array}\right] \sim \left[\begin{array}{cc|c} 1 & 0 & -12 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \end{array}\right]$$

So $(c_1, c_2) = (-12, 2)$